Predictive Maintenance Report

Multilabel Failure Classification with Machine Learning

AI4I 2020 Predictive Maintenance Dataset · UCI Machine Learning Repository

Danilo Luvizotto Gonçalves
13/03/2026

Table of Contents

01 · Business Problem

$50B+

Annual cost of
unplanned downtime
(manufacturing, global)

30–40%

Cost reduction with
predictive vs. reactive
maintenance

~97%

Operational time
is failure-free
(rare event challenge)

The Challenge

Industrial equipment fails in multiple, simultaneous ways. Each failure type (tool wear, heat dissipation, power failure, overstrain, random) requires a distinct maintenance action. Traditional binary alarms miss this granularity — and extreme class imbalance (~1–3% failure rate per type) makes the problem technically demanding.

To maintain business competitivity, our industry need a solution that handles these failures with precision, to minimize costs.

02 · Dataset Overview — AI4I 2020 (UCI)

10,000
Samples

6 predictors
Features

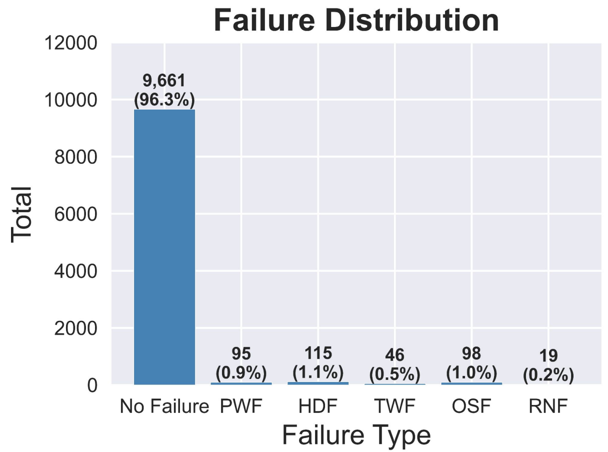
5 failure types
Target labels

≈ 1% per type
Positive rate

Input Features

- Air temperature [K]
- Process temperature [K]
- Rotational speed [rpm]
- Torque [Nm]
- Tool wear [min]
- Product type (L / M / H)

Failure Type Distribution



Failure Types:

Overstrain Failure (OSF)
Power Failure (PWF)
Heat Dissipation Failure (HDF)
Tool Wear Failure (TWF)
Random Failure (RNF)

03 · Methodology

Data Ingestion

AI4I 2020
UCI

EDA & Preprocessing

Scaling
Encoding

Imbalance Handling

RandomOver-Sampler

Hyperparameter Tuning

Optuna

Model Training

Binary Testing:
Logistic Regression
Random Forests
XGBoost

Multilabel Testing:
Random Forests
XGBoost

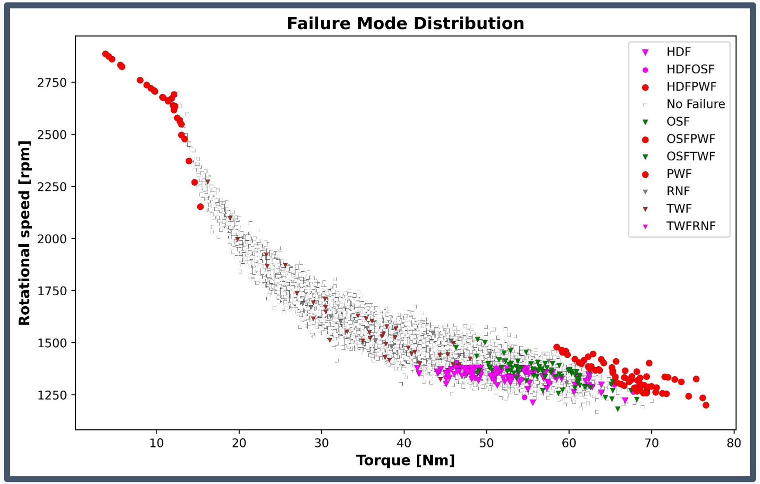
Evaluation

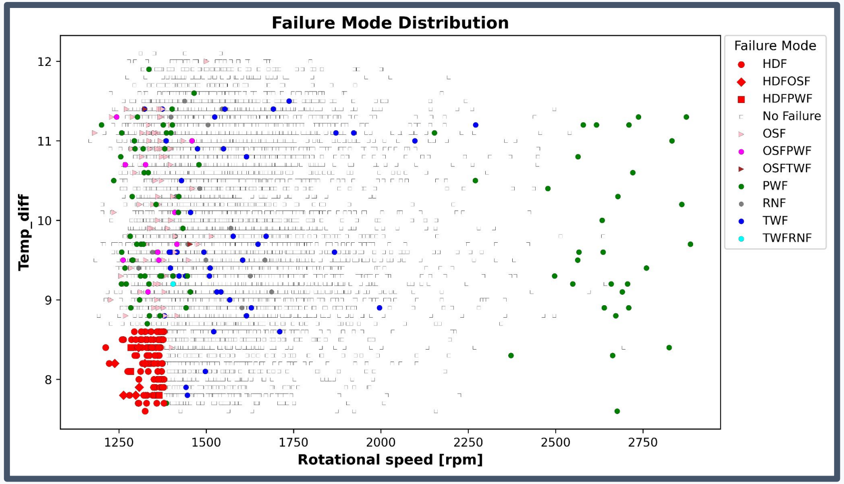
AUC-PR
MCC · F1

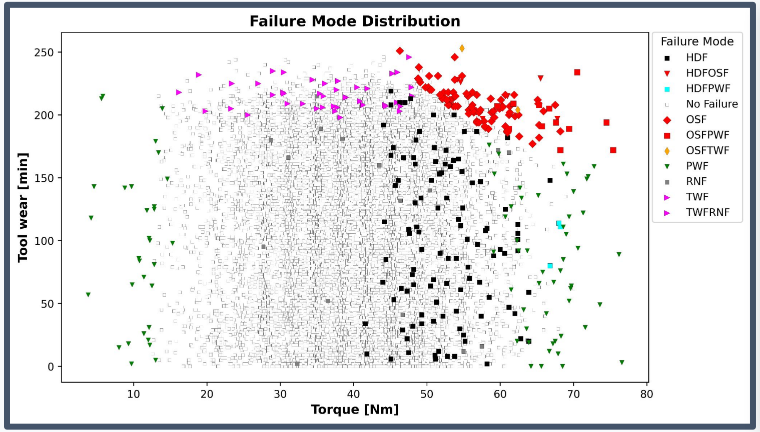
Key Design Decisions

- ✓ Oversampling applied INSIDE each CV fold to prevent data leakage
- ✓ Optuna optimizes MCC as objective — robust to severe imbalance
- ✓ Stratified K-Fold (k=5) with multilabel compatible splitting
- ✓ Feature Engineering for model simplification

04 · Data Visualization - Preliminary Insights







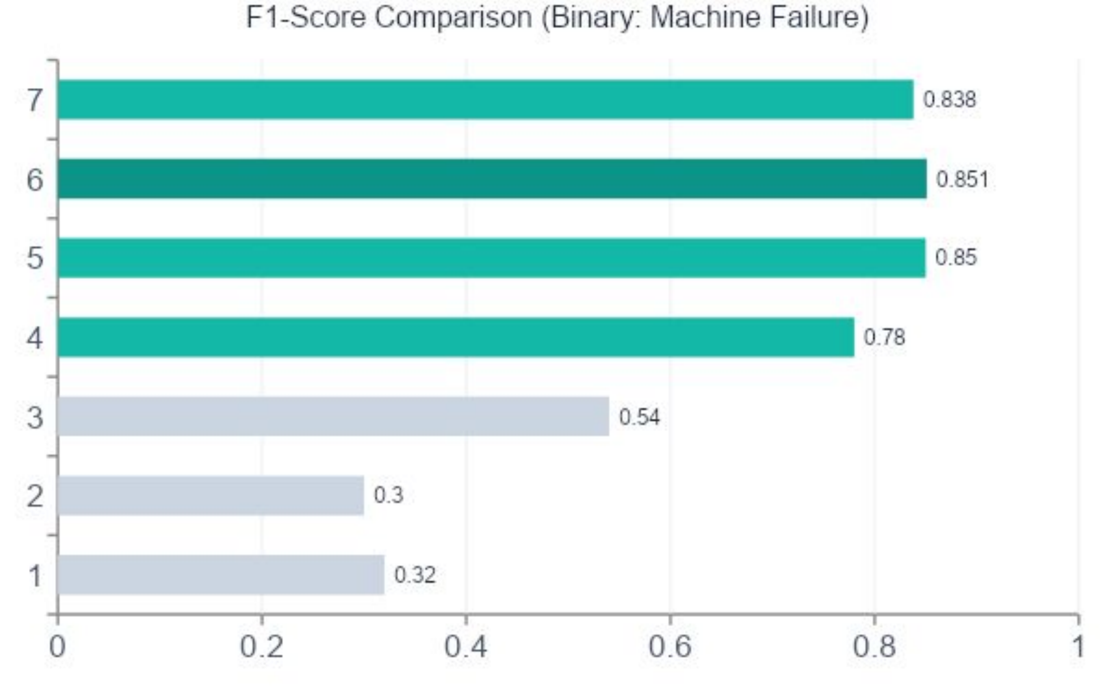
Key Insights

Some Failure modes showed a clear pattern (OSF, PWF and HDF) in the preliminary analysis, indicating that our models were more likely to correctly predict them, while TWF and RNF patterns were not clearly seen through the plots made. That indeed reflected into poorer performance in predicting these kind of failures.

05 · Results & Model Comparison - Binary Model

🏆 **Best Model (Binary): RFC_Reduced + Optuna** · F1 = 0.851 · MCC (CV) = 0.796 · Precision = 0.864 · Recall = 0.838

57 Correct Failure Predictions out of 68 - 11 false negatives and 9 false positives



Key Metrics Summary

Model	MCC (CV)	F1
LR Baseline	0.395	0.32
LR Optimized	0.459	0.54
RFC Baseline	—	0.78
RFC Reduced ★	**0.796**	**0.851**
XGBoost	—	0.838

05 · Results & Model Comparison - Multilabel Model

🏆 **Best Model (Binary):** **RFC_Single + Optuna** · Total correct predictions: 52, out of 73. RNF and TWF are not predicted by our model. **Considering only HDF, PWR and OSF → 52 out of 62 - 84% of correct predictions (only 2 FP)**

```
Prediction performance for RFC_Single

              precision    recall  f1-score   support

         TWF       0.00      0.00      0.00         9
         HDF       0.92      1.00      0.96        23
         PWF       1.00      0.68      0.81        19
         OSF       1.00      0.80      0.89        20
         RNF       0.00      0.00      0.00         4

   micro avg       0.96      0.69      0.81        75
   macro avg       0.58      0.50      0.53        75
weighted avg       0.80      0.69      0.74        75
 samples avg       0.02      0.02      0.02        75
```

```
Sum of correct predictions for RFC_Single: 52
Sum of correct False Positives for RFC_Single: 2
Sum of correct False Negatives for RFC_Single: 10
```

Key Metrics Summary

Failure type	Detected failures	FN+FP
HDR	23	0+2
OSF	16	4+0
PWF	13	6+0
TWF	**0**	**9+0**
RNF	0	4+0

06 · Key Insights — SHAP Feature Importance

Rotational Speed #1 predictor

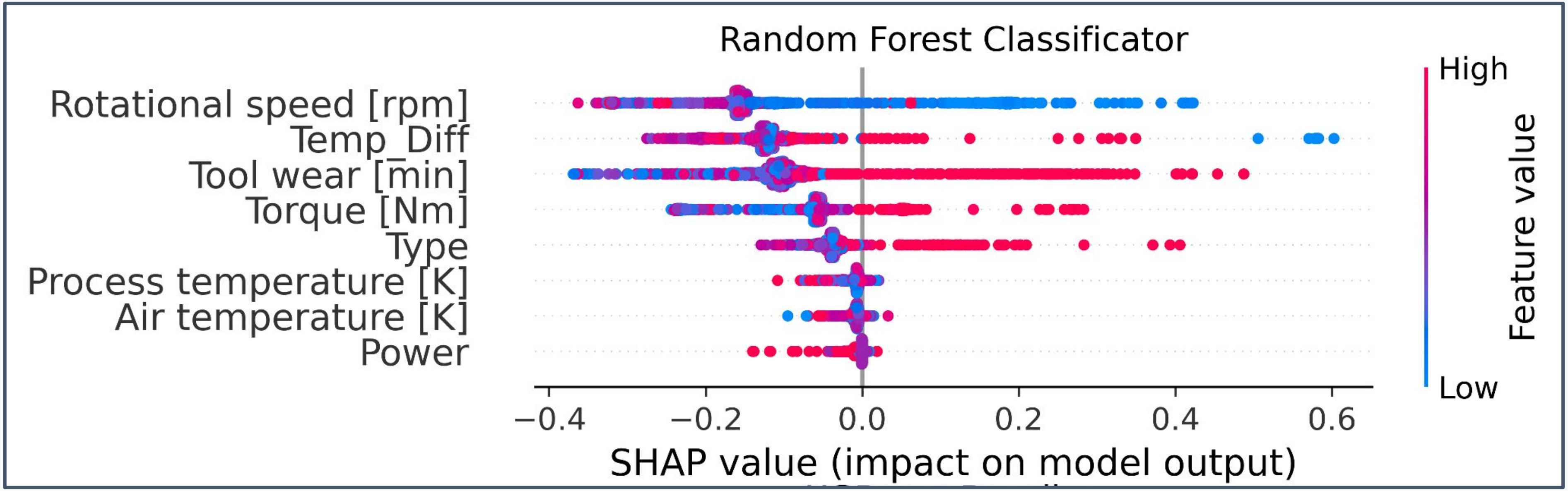
SHAP analysis on RFC_Reduced shows Rotational speed has the highest mean |SHAP| — Rotational speed category drives failure risk more than any sensor reading.

Torque & Tool Wear are key sensors as well

Tool wear 3rd, torque 4th. Together these three mechanical signals capture the bulk of failure risk.

Engineered Temp_Diff matters

The engineered feature Temp_Diff (process − air temp) outperforms the raw individual temperatures, confirming its predictive value for heat dissipation failures.



07 · Conclusions & Next Steps

- RFC_Reduced + Optuna achieves F1 = 0.851 and MCC (CV) = 0.796 for binary machine failure, outperforming all baselines
  57 correct predictions, 9 false positives and 11 false negatives → **74% of total machine downtime would be predictive**

- Multilabel RFC_Single correctly identifies 52 of 62 positive cases (2 FP, 10 FN) across 3 failure types on the test set

  84% of Power Failures, Heat Dissipation Failures and Overstrain Failures would be held by predictive maintenance

- Oversampling strictly within CV folds prevents data leakage — imblearn Pipeline ensures unbiased performance estimates

- TWF and RNF remain hard to detect (recall 0%) — extremely rare events with fewer than 50 test samples require dedicated strategies

Next Steps

· LightGBM integration · Real-time inference pipeline · Deep learning baseline (LSTM)

APPENDIX

Technical Details · Hyperparameters · Extended Metrics · Cross-Validation

Appendix A · Optuna Hyperparameter Search Space - Binary Model

Logistic Regression

C (regularization)	**1e-3 → 10**
Penalty	**l1, l2**
Solver	**liblinear, saga**
Max iter	**100 → 500**

Random Forest

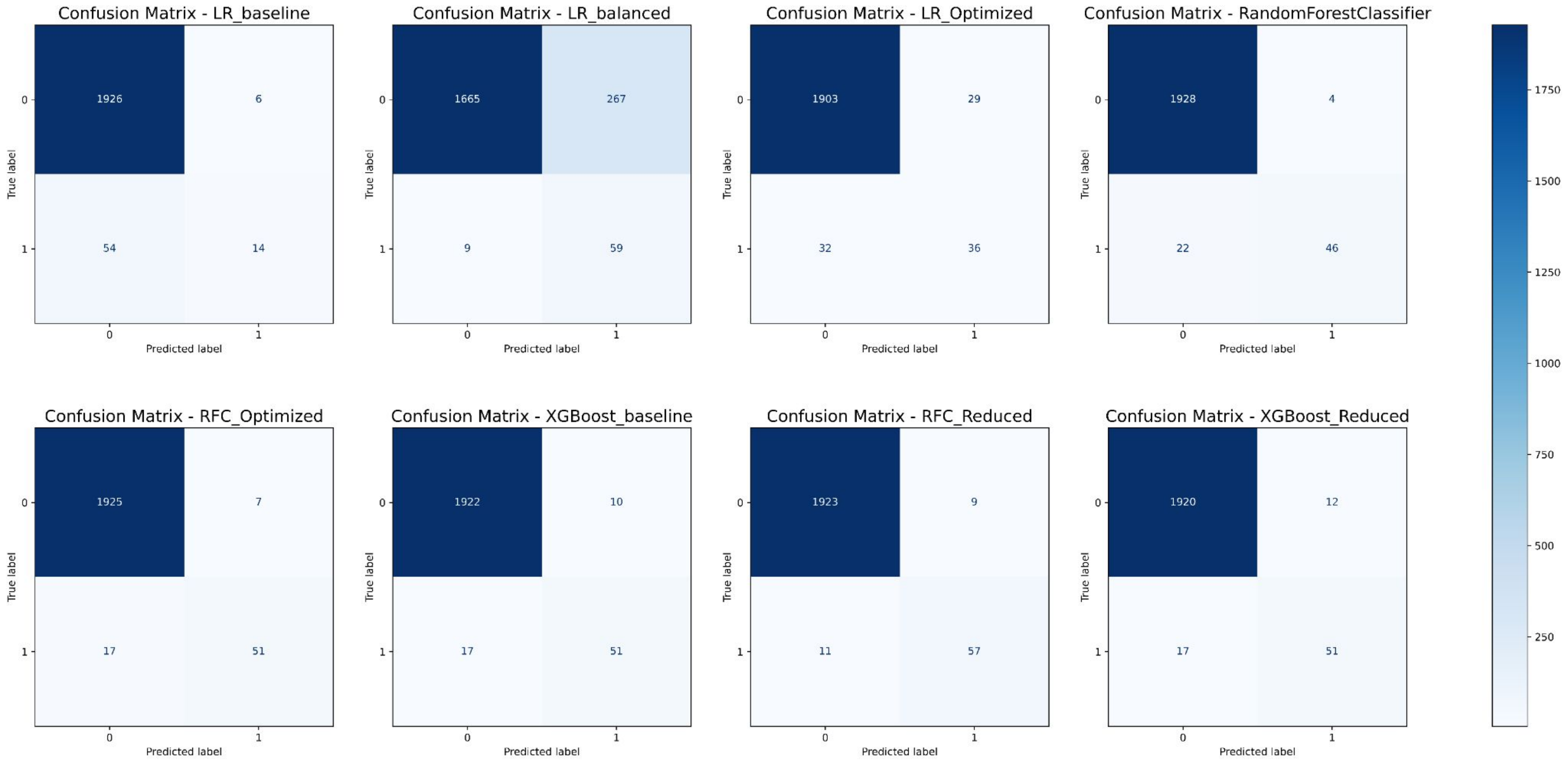
n_estimators	**50 → 500**
max_depth	**3 → 20**
min_samples_split	**2 → 20**
max_features	**sqrt, log2**

XGBoost

n_estimators	**50 → 500**
max_depth	**3 → 10**
learning_rate	**0.01 → 0.3**
subsample	**0.5 → 1.0**
colsample_bytree	**0.5 → 1.0**
scale_pos_weight	**auto**

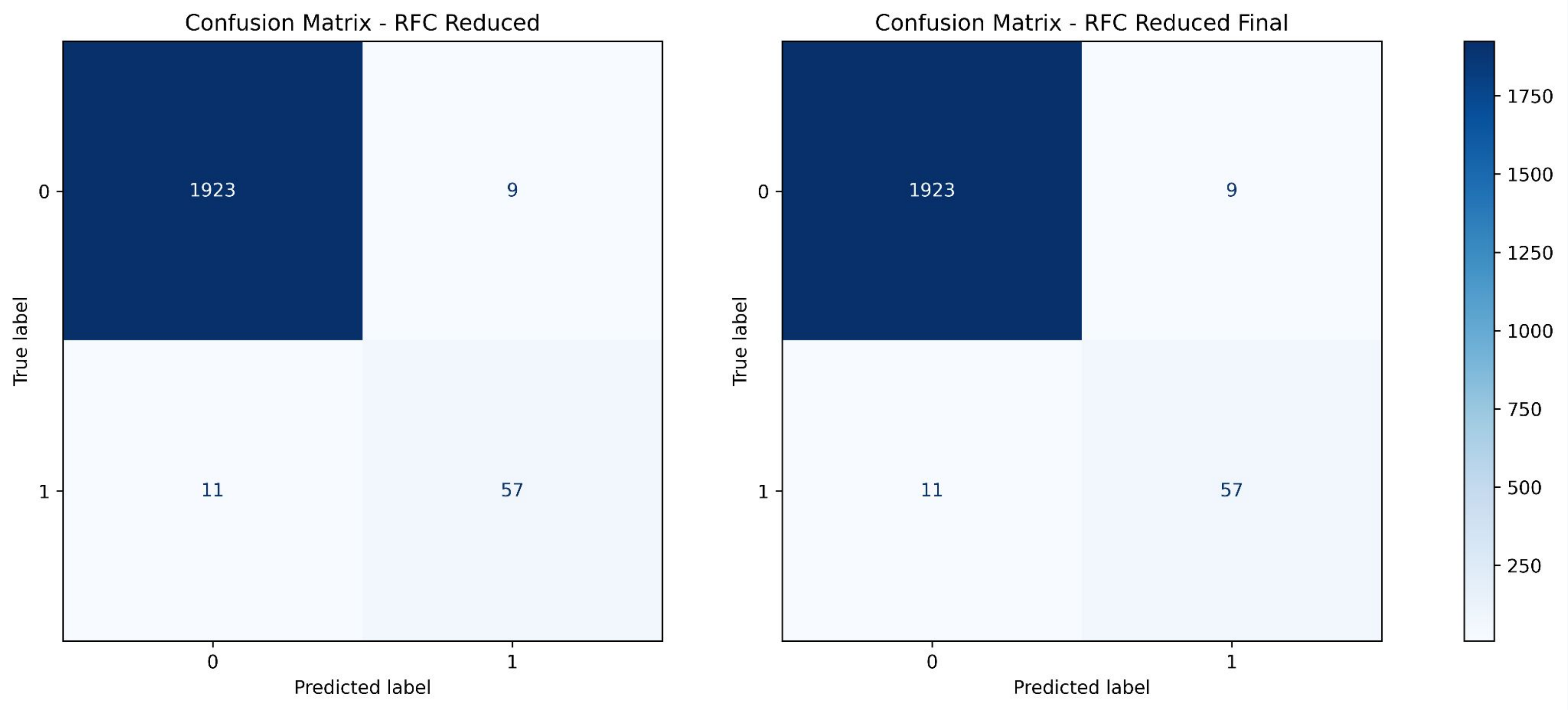
Objective: maximize AUC-PR · 50 trials per model per label · Sampler: TPESampler · Pruner: MedianPruner

Appendix B · Per-Label Confusion Matrix Summary — RFC_Reduced



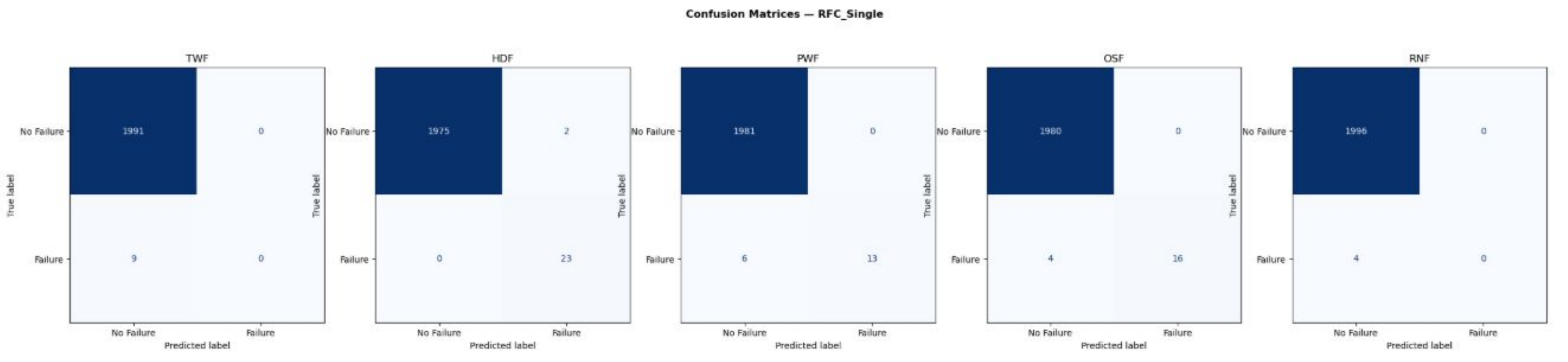
Test set results (20% split, 2000 samples). TWF & RNF have very few positive test cases (≤9), making recall 0% despite model detecting some in training. Multilabel Optuna best F1-micro = 0.8863.

Appendix B · Per-Label Confusion Matrix Summary — RFC_Reduced



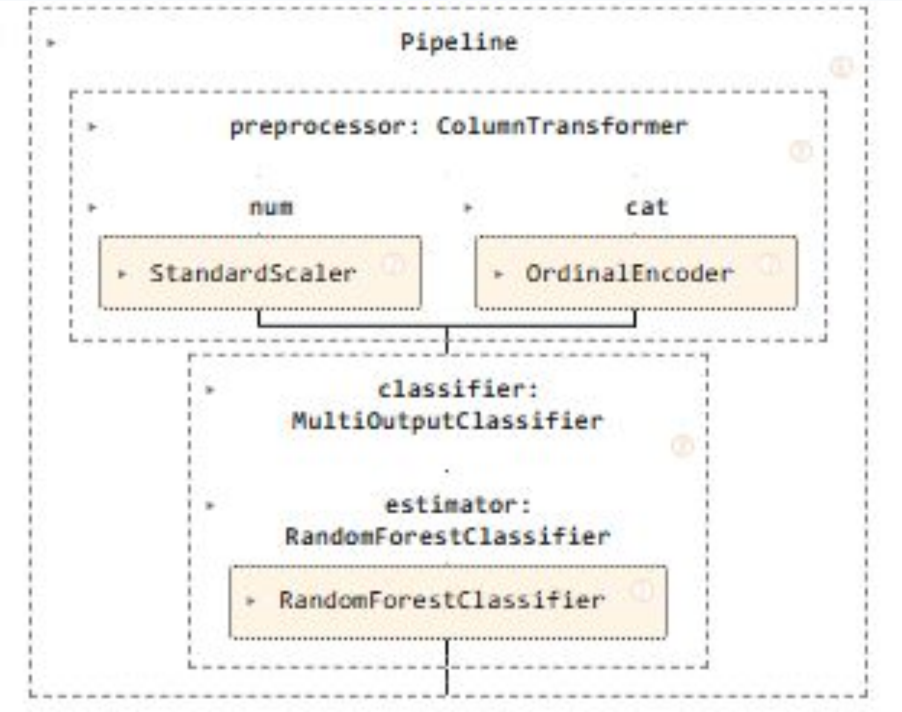
Test set results (20% split, 2000 samples). TWF & RNF have very few positive test cases (≤9), making recall 0% despite model detecting some in training. Multilabel Optuna best F1-micro = 0.8863.

Appendix B · Per-Label Confusion Matrix Summary — RFC_Single (Multilabel)



Test set results (20% split, 2000 samples). TWF & RNF have very few positive test cases (≤9), making recall 0% despite model detecting some in training. Multilabel Optuna best F1-micro = 0.8863.

Appendix B · RFC_Single - Pipeline and parameters used post tuning



estimator: RandomForestClassifier

RandomForestClassifier

▼ Parameters

n_estimators	116
criterion	'log_loss'
max_depth	None
min_samples_split	8
min_samples_leaf	1
min_weight_fraction_leaf	0.0
max_features	'sqrt'
max_leaf_nodes	None
min_impurity_decrease	0.0
bootstrap	True
oob_score	False
n_jobs	None
random_state	42
verbose	0
warm_start	False
class_weight	'balanced'
ccp_alpha	0.0008972512738601409
max_samples	None
monotonic_cst	None

Test set results (20% split, 2000 samples). TWF & RNF have very few positive test cases (≤9), making recall 0% despite model detecting some in training. Multilabel Optuna best F1-micro = 0.8863.

Appendix C · Cross-Validation Strategy & Leakage Prevention

Stratified 5-Fold Cross-Validation with Oversampling Pipeline

Fold 1	Val	Train+OS	Train+OS	Train+OS	Train+OS
Fold 2	Train+OS	Val	Train+OS	Train+OS	Train+OS
Fold 3	Train+OS	Train+OS	Val	Train+OS	Train+OS
Fold 4	Train+OS	Train+OS	Train+OS	Val	Train+OS
Fold 5	Train+OS	Train+OS	Train+OS	Train+OS	Val

Training + Oversampling (inside fold)

Validation (original distribution)

Why it matters

Applying oversampling before CV splits allows synthetic samples from the training set to appear in validation — inflating performance.

Using imblearn Pipeline ensures the oversampler sees only each fold's training split, giving honest estimates.